



Network type sp^3 boron-based single-ion conducting polymer electrolytes for lithium ion batteries

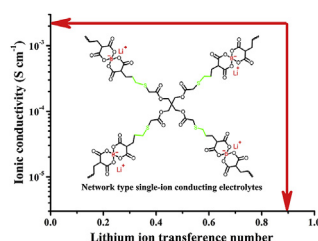
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HIGHLIGHTS

- Boron-based single-ion conducting gel polymer electrolytes are designed.
- The electrolytes are fabricated by facile in situ thiol-ene click reaction.
- Ionic conductivity reaches $1.47 \times 10^{-3} \text{ S cm}^{-1}$ at 25°C .
- High lithium transference number of 0.89 is obtained.
- The LiFePO_4 cell delivers a capacity of 124 mA h g^{-1} at 1 C rate after 500 cycles.

GRAPHICAL ABSTRACT



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ABSTRACT

Electrolytes play a vital role in modulating lithium ion battery performance. An outstanding electrolyte should possess both high ionic conductivity and unity lithium ion transference number. Here, we present a facile method to fabricate a network type sp^3 boron-based single-ion conducting polymer electrolyte (SIPE) with high ionic conductivity and lithium ion transference number approaching unity. The SIPE was synthesized by coupling of lithium bis(allylmalonato)borate (LiBAMB) and pentaerythritol tetrakis(2-mercaptoacetate) (PETMP) via one-step photoinitiated in situ thiol-ene click reaction in plasticizers. Influence of kinds and content of plasticizers was investigated and the optimized electrolytes show both outstanding ionic conductivity ($1.47 \times 10^{-3} \text{ S cm}^{-1}$ at 25°C) and high lithium transference number of 0.89. This ionic conductivity is among the highest ionic conductivity exhibited by SIPEs reported to date. Its electrochemical stability window is up to 5.2 V. More importantly, $\text{Li}/\text{LiFePO}_4$ cells with the prepared single-ion conducting electrolytes as the electrolyte as well as the separator display highly reversible capacity and excellent rate capacity under room temperature. It also demonstrates excellent long-term stability and reliability as it maintains capacity of 124 mA h g^{-1} at 1 C rate even after 500 cycles without obvious decay.

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1. Introduction

Rapid development of portable electronic devices, smart grids, electric vehicles (EVs) and hybrid electric vehicles (HEVs) demand secondary batteries with higher power density, enhanced safety and lower cost [1,2]. Lithium ion batteries are considered to be one of the most promising energy storage devices for these applications

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due to their lightweight and high energy density [3]. Electrolytes that conduct electricity by ions between electrodes play a vital role in modulating the overall lithium ion battery performance [4–6]. It is essential to develop electrolytes with high ionic conductivity, high lithium ion transference number, broad electrochemical window and good safety, which has been a subject of intense research efforts [7–9]. Typical commercial liquid electrolytes consist of lithium salts and flammable organic solvents. However, the high flammability, potential leakage, the inherent concentration polarization and formation of dendrite responsible for explosion hazards have raised serious concerns on the battery safety and performance [10–12].

Polymer electrolytes consisting of a polar polymer matrix and lithium salts appear to be a promising candidate to liquid electrolytes, due to their low flammability, no liquid leakage and flexibility in the design of the cells [13–16]. While almost all liquid electrolytes and polymer electrolytes used in both commercial lithium-ion batteries and prototype are so-called bi-ionic conductors [17,18]. In bi-ionic conductors both lithium ions and anions migrate during charging and discharging process [19]. Anions move 5–10 times faster than lithium ions and contribute more to the ionic conductivity than lithium ions. But there is no electrode reaction for the anions. As a result, lithium ion transference number is far below the value of 0.5 and the buildup of the anions at the electrode/electrolyte interface causes concentration polarization [18,19]. It leads to serious decay of conductivity that limits the cell life, charging rate and energy density [20–22].

Single-ion conducting polymer electrolytes (SIPes) were proposed to solve the issues on dual-ion movement [23,24]. In this approach, the anions are covalently tethered onto a rigid polymer chain and the counter lithium ions are the only mobile species. By restricting the anionic movement, SIPes displayed lithium ion transference number approaching unity [20,25]. However, such anions immobilization always results in a drop of ionic conductivity for solid-state single-ion conducting electrolytes (typically below 10^{-7} S cm $^{-1}$ at ambient temperature) [26–32]. Efforts such as introduction of weakly coordinating anionic species [33–35] and optimization of the macromolecular architecture to lower glass transition temperature and increase local segmental motion [36–38] have been devoted to improve ionic conductivity. So far, few solid-state SIPes have showed satisfactory ionic conductivity and battery performance [18]. Most solid-state SIPes enable batteries to operate only at high temperatures with low charge–discharge rates [20,37–40]. For example, Armand et al. [20] reported PSTFSILi-PEO-PSTFSILi triblock copolymers with conductivity of 1.3×10^{-5} S cm $^{-1}$ at 60 °C. But ionic conductivity is lower than 10^{-6} S cm $^{-1}$ at 45 °C and battery prototypes based on these electrolytes have been run only at moderately high temperature of 60–80 °C. Gerbaldi and coworkers [38] described single-ion block copoly(ionic liquid)s showing low T_g (up to –61 °C) and ionic conductivity of 2.3×10^{-6} S cm $^{-1}$ at 25 °C in a dry state. Yet Li metal/LiFePO $_4$ cells utilizing these electrolytes cycled at 70 °C with relatively low charge–discharge rates. Novel polymeric lithium borate salts were developed by Prof. Angell, among which the highest ionic conductivity is 10^{-5} S cm $^{-1}$ at room temperature [41,42].

The concept of single-ion conducting gel polymer electrolyte was recently proposed to enhance ionic conductivity at ambient temperature [43–45]. Single-ion conducting gel polymer electrolytes can attain high ionic conductivity with the addition of plasticizers while in conventional solid-state SIPes ion conduction is realized via a hopping mechanism [43,46]. Single-ion conducting gel polymer electrolytes display both high lithium transference number (approaching unity) and high ionic conductivity (up to 10^{-3} S cm $^{-1}$) at ambient temperature [19,21,25,46–53]. Wang et al.

[19] prepared single-ion electrolytes based on aromatic poly(arylene ether)s with pendant lithium perfluoroethyl sulfonates. They showed high conductivity ($>10^{-3}$ S cm $^{-1}$ at 25 °C) and lithium transference number when saturated with organic carbonates. The assembled cell performed well at ambient temperature. Cheng and coworkers [47] reported a dense transparent polymeric single-ion conductor with ionic conductivity of 0.104 mS cm $^{-1}$ at 25 °C and lithium transference number of 0.92. Cui et al. [54] prepared polymeric lithium borate salts PLPB@PVDF-HFP with favorable ionic conductivity of 1.8×10^{-4} S cm $^{-1}$ at 30 °C. However, in these reports the electrolyte preparation processes are complicated or potentially expensive for practical applications to some degree [50].

In this paper, we present a facile method to prepare single-ion conducting polymer electrolytes via one-step photoinitiated in situ thiol-ene click reaction between lithium bis(allylmalonate) borate (LiBAMB) and pentaerythritol tetrakis(2-mercaptoacetate) (PETMP) in plasticizers such as gamma-butyrolactone (GBL), propylene carbonate (PC) and ethylene carbonate (EC). Due to the weak coordination between lithium ions and charge delocalized sp^3 boron anions as well as good compatibility between the single-ion conducting polymer and the plasticizers, the polymer electrolytes show excellent ionic conductivity ($>10^{-3}$ S cm $^{-1}$ at 25 °C) and high lithium transference number. More importantly, Li/LiFePO $_4$ cell with the prepared SIPes as the electrolyte as well as the separator displays highly reversible capacity and excellent rate capacity even up to 5C rate. It also demonstrates excellent long-term stability and reliability as it maintains capacity of 124 mA h g $^{-1}$ at 1 C rate even after 500 cycles without obvious decay.

2. Experimental

2.1. Materials

Allylmalonic acid (Alfa-Aesar), lithium carbonate (Aladdin), boric acid (Aladdin), 2,2-dimethoxy-2-phenylacetophenone (DMPA) (Aladdin), propylene carbonate (PC) (Sigma-Aldrich), gamma-butyrolactone (GBL) (Aladdin), ethylene carbonate (EC) (Sigma-Aldrich), *N*-methyl-2-pyrrolidone (NMP) (Aladdin), super P (TIMCAL), poly(vinylidene fluoride) (PVDF) (Arkema) and LiFePO $_4$ (MTI Kejing Co., Ltd) were used as-purchased without further purification. Pentaerythritol tetrakis(2-mercaptoacetate) (PETMP) (Sigma-Aldrich) was dried and stored with 4A molecule sieves in a glove box. Acetonitrile were dried with molecular sieves before use.

2.2. Synthesis of lithium bis(allylmalonate)borate (LiBAMB)

Allylmalonic acid (60 mmol), lithium carbonate (15 mmol) and boric acid (30 mmol) were added in 150 mL dry acetonitrile and heated under nitrogen gas flow in an 80 °C oil bath for 12 h. After cooling down, it was filtered and the solvent was removed under reduced pressure. White solid was obtained after drying under high vacuum at 60 °C for 48 h with a yield of 92.4%. It was quickly transferred into an argon filled glove box (H $_2$ O and O $_2$ content <0.1 ppm) and stored in it. ^1H NMR (DMSO, ppm): 5.76 (m, 1H), 5.03 (d, 1H), 4.97 (d, 1H), 3.54 (t, 1H), 2.55 (t, 2H). ^{13}C NMR (DMSO, ppm): 168.18, 135.64, 117.48, 47.89, 30.92. ^{11}B NMR (DMSO, ppm): 2.57.

2.3. Preparation procedure of LiBAMB-PETMP SIPes

In an argon filled glove box, LiBAMB (2.5 mmol), PETMP (1.25 mmol) and DMPA (0.25 mmol) were dissolved in 5 mL GBL and then poured on PTFE dishes. It was cured under UV light (365 nm) for 30 min. After that, a self-standing, flexible and transparent film was formed. The film was punched into circular

rounds and stored in an argon-filled glove box for electrochemical measurements and battery test.

2.4. Characterization of LiBAMB and LiBAMB-PETMP

^1H NMR spectra, ^{13}C NMR spectra and ^{11}B NMR spectra were taken on Bruker DRX-500 NMR spectrometer with dimethyl sulfoxide- d_6 as solvent and tetramethylsilane as internal standard. Fourier-transform infrared (FT-IR) spectra were measured on a Perkin-Elmer Spectrum 100 Fourier transform spectrometer.

2.5. Ionic conductivity measurement

Ionic conductivity was measured by electrochemical impedance spectroscopy (EIS) using the Solartron 1255B frequency response analyzer with the EIS module over a frequency range of 1 Hz to 1 MHz and an AC excitation voltage of 10 mV from room temperature to 90 °C with an increment of 10 °C. The cell configuration for ionic conductivity tests was “SS|film|SS”, in which the polymer electrolytes film was sandwiched between two stainless steel electrode discs and sealed in a standard coin cells (CR2032) inside a dry argon filled glove box. The ionic conductivity was calculated using the following equation:

$$\sigma = \frac{l}{RA}$$

where l is the film thickness, R stands for the bulk resistance, and A represents the area.

2.6. Lithium ion transference number measurement

Lithium ion transference number was measured by the method based on the combination of chronoamperometry (CA) and electrochemical impedance spectroscopy (EIS) suggested by Bruce and coworkers [55]. The cell configuration was “Li|film|Li”. EIS was performed from 1 Hz to 1 MHz with an AC excitation voltage of 10 mV before and after the CA test. CA was conducted with a static potential polarization of 10 mV. The transference number (T_+) was calculated as:

$$T_+ = \frac{I_s(\Delta V - I_0 R_0)}{I_0(\Delta V - I_s R_s)}$$

where ΔV corresponds to the potential applied on the cell, I_0 and I_s are the initial current and the steady state current, and R_0 and R_s represent the initial resistance and the steady state resistance of the passivation layers on the lithium electrode, respectively.

2.7. Electrochemical stability window

Electrochemical stability window was measured by electrochemical linear sweep voltammetry (LSV) with the cell configuration of “SS|film|Li”. The measurements were conducted from −1 V and 6.5 V (vs. Li^+/Li) at a scan rate of 1 mV s^{-1} .

2.8. Battery assembly and measurement

LiFePO_4 , super P and PVDF at the ratio of 80:10:10 by weight were added in NMP and magnetic stirred to form a well-dispersed slurry, which was coated on a flat aluminum foil by the doctor blade process. After drying in the oven, the coated aluminum foil was cut into circular rounds and further dried at 50 °C under high vacuum for at least 24 h. The loading of LiFePO_4 is around 1.7 mg cm^{-2} . 2032-type Li/LiFePO₄ coin cells were assembled with the single-ion

conducting electrolytes film as the separators and electrolyte. Charge and discharge tests for the batteries were performed at 25 °C with the upper cut-off voltage of 4.0 V and the lower cut-off voltage of 2.5 V.

3. Results and discussions

3.1. Preparation of monomers and network type sp^3 boron-based single-ion conducting electrolytes

The monomer, lithium bis(allylmalonato)borate (LiBAMB) was synthesized via one-step simple reaction of allylmalonic acid, lithium carbonate and boric acid. Successful synthesis of LiBAMB was confirmed by the ^1H NMR (Fig. S1), ^{13}C NMR (Fig. S2) and ^{11}B NMR (Fig. 1) characterizations, which agrees well with the work of Sun and coworkers [44]. The ^{11}B NMR spectrum (Fig. 1) exhibits only a strong peak at 2.57 ppm, reflecting a typical sp^3 hybridized state of boron atoms [56]. No other peaks are observed in the ^{11}B NMR spectrum, indicating high purity of the prepared LiBAMB.

Network type sp^3 boron-based single-ion conducting gel polymer electrolytes (LiBAMB-PETMP) were prepared via one-step photoinitiated in situ thiol-ene click reaction between LiBAMB and pentaerythritol tetrakis(2-mercaptoacetate) (PETMP) in plasticizers such as gamma-butyrolactone (GBL), propylene carbonate (PC) and ethylene carbonate (EC) (Scheme 1). The prepared plasticized LiBAMB-PETMP films appear to be self-standing, flexible and transparent as showed in Fig. 2. To investigate the mechanical properties, nanoindentation was performed and the evolution of Young's modulus E versus the indentation depth h of LiBAMB-PETMP/85 wt. % GBL is shown in Fig. S3. Young's modulus of the gel polymer electrolyte obtained from the constant value of the curve is about 0.092 GPa.

The FT-IR spectra of LiBAMB and LiBAMB-PETMP are shown in Fig. 3. For LiBAMB, the strong peak at 1710 cm^{-1} corresponds to the $\text{C}=\text{O}$ bonds of the carboxylic group. The absorption peaks of the allyl group appear at 1641 cm^{-1} ($\nu_{\text{C}=\text{C}}$, $-\text{CH}=\text{CH}_2$) and 3091 cm^{-1} ($\nu_{\text{C-H}}$, $=\text{CH}_2$). While for LiBAMB-PETMP, peaks at 1641 cm^{-1} and 3091 cm^{-1} are not observed, indicating complete conversion of the allyl group.

3.2. Ionic conductivity

Ionic conductivity of the network type sp^3 boron-based SIPEs (LiBAMB-PETMP) was measured by EIS method. First of all, influence of different solvents as plasticizers was investigated. As shown in Table 1, when with identical 66 wt. % content of plasticizers, single-ion conducting electrolytes with GBL (LiBAMB-PETMP/GBL)

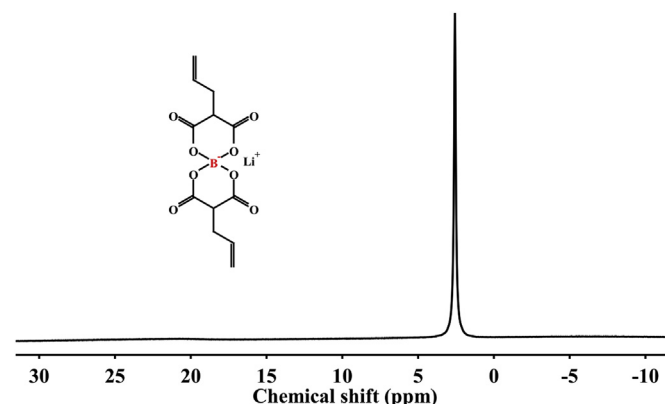
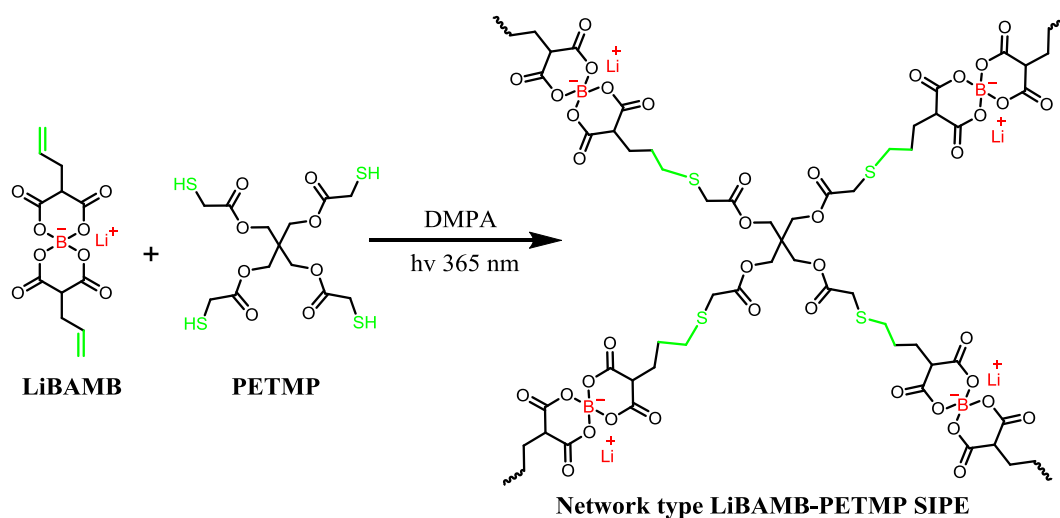


Fig. 1. ^{11}B NMR spectrum of LiBAMB ($\text{DMSO-}d_6$ as solvent).



Scheme 1. Synthesis of LiBAMB-PETMP via one-step photoinitiated thiol-ene click reaction.

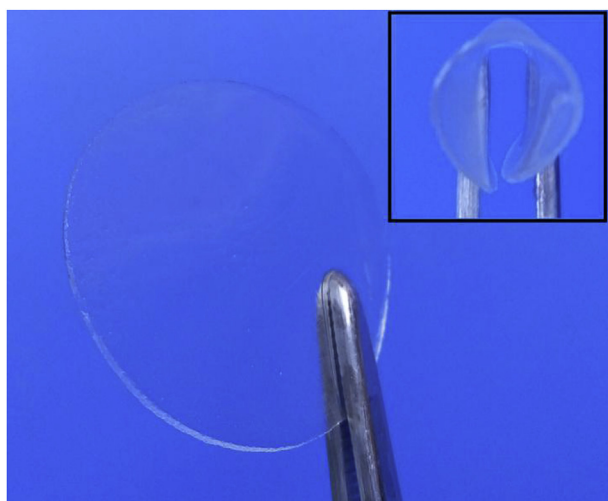


Fig. 2. Photograph of the prepared LiBAMB-PETMP/85 wt. % GBL film.

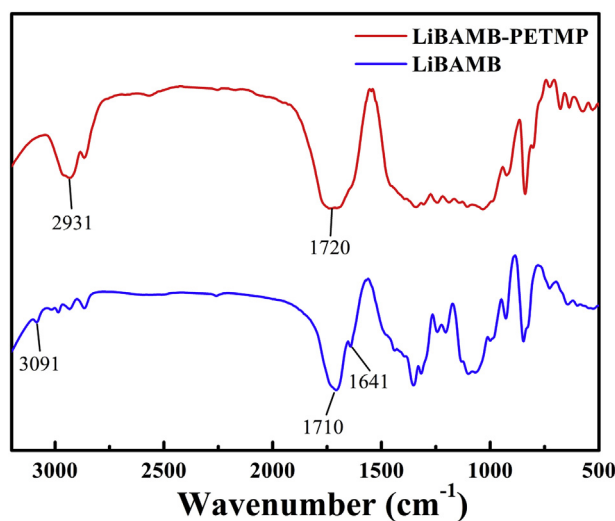


Fig. 3. FT-IR spectra of LiBAMB-PETMP and LiBAMB.

Table 1

Comparison of different solvents as plasticizers.

Solvent	Content (wt. %)	Ionic conductivity (S cm^{-1}) (25 °C)
GBL	66	3.51×10^{-4}
PC	66	1.00×10^{-4}
PC + EC (1:1 vol)	66	0.97×10^{-4}
GBL + PC (1:1 vol)	66	2.15×10^{-4}
GBL + EC (1:1 vol)	66	2.47×10^{-4}

as plasticizer shows the highest ionic conductivity of $3.51 \times 10^{-4} \text{ S cm}^{-1}$ at room temperature, while with cyclic carbonate (PC, PC + EC) as plasticizer, the ionic conductivity is only around $1.00 \times 10^{-4} \text{ S cm}^{-1}$. The electrolyte plasticized by GBL and PC (or EC) mixture showed moderate ionic conductivity. GBL is a sort of cyclic carboxylate once used in primary lithium batteries, of which melting point is -43.53°C and boiling point is 204°C . Researches have reported that lithium borate salts-GBL electrolyte exhibits good electrochemical stability and GBL as the solvent of the lithium borate salts can increase the solubility and conductivity dramatically [57,58], which is in consistence with our experimental results. Therefore, GBL was chosen as plasticizer for further investigation of the influence of content of plasticizer on the ionic conductivity of the single-ion conducting gel polymer electrolyte.

Ionic conductivity of the network type sp^3 boron-based SIPEs (LiBAMB-PETMP/GBL) with different content of GBL from room temperature to 90°C with an increment of 10°C are shown in Fig. 4. For all LiBAMB-PETMP/GBL SIPEs, the ionic conductivity increases with increasing temperature. The isothermal ionic conductivity of LiBAMB-PETMP/GBL displays an obvious maximum against the content of GBL. LiBAMB-PETMP/GBL containing 85 wt. % of GBL exhibits the highest ionic conductivity of approximately $1.47 \times 10^{-3} \text{ S cm}^{-1}$ at 25°C , which is among the highest ionic conductivity exhibited by single-ion conducting gel polymer electrolytes reported to date as summarized in Table 2. This value is also higher than those of traditional liquid electrolytes incorporated with polyolefin separators, that is, $0.1\text{--}1 \times 10^{-3} \text{ S cm}^{-1}$ [4,19]. For instance, ionic conductivity of 1 M LiPF_6 in DMC + EC using Celgard 2500 as separator was reported to be $0.85 \times 10^{-3} \text{ S cm}^{-1}$ at room temperature [4]. Such high ionic conductivity is due to weak co-ordination between lithium ions and sp^3 boron anions that are bonded with electron withdrawing groups. Furthermore, the addition of GBL plasticizer enhances lithium ions dissociation and

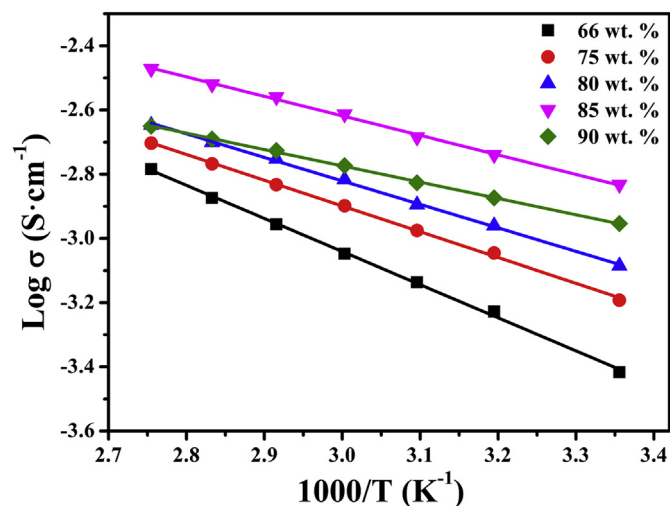


Fig. 4. Temperature dependence of ionic conductivity for LiBAMB-PETMP/GBL SIPEs with various GBL contents.

the mobility of the lithium ion in the polymer matrix. Fits to the Arrhenius equation are shown in Fig. 4 as lines, and the corresponding parameters are also given in Table 3. The temperature dependence of the conductivity of LiBAMB-PETMP/GBL follows obviously an Arrhenius behavior. Ionic conductivity increases with increasing GBL content from 66 wt. % to 85 wt. %, being in accordance with the reduction of activation energy E_a , which is attributed to the increase of the mobility of the lithium ion within the

Table 3

Arrhenius fitting parameters of ionic conductivities for LiBAMB-PETMP/GBL SIPEs.

GBL content (wt. %)	E_a (kJ/mol)	σ_0 (S cm ⁻¹)
66	8.58	1.06
75	6.68	0.61
80	6.09	0.54
85	5.05	0.45
90	4.23	0.29

polymer matrix. However, ionic conductivity drops a slightly when GBL content increases from 85 wt. % to 90 wt. %. Such behavior can be reasonably explained by the decrease in concentration of lithium ion in the electrolyte because the concentration of lithium ion decreases from 0.34 M to 0.25 M when GBL content increases from 85 wt. % to 90 wt. %.

3.3. Lithium ion transference number

The single-ion conducting feature of LiBAMB-PETMP/GBL can be confirmed by lithium ion transference number measurement. Determination of lithium ion transference number for the developed LiBAMB-PETMP/85 wt. % GBL SIPEs was performed through the combination of chronoamperometry (CA) and electrochemical impedance spectroscopy (EIS) as shown in Fig. 5. The T^+ value was calculated to be 0.89, which infers the single-ion conducting behavior of the LiBAMB-PETMP/GBL SIPEs as presumed. The significantly higher value of T^+ value accounts for the restricted movement of anions that is part of crosslinking network.

Table 2

Summary of current single-ion conducting gel polymer electrolytes.

Electrolyte	Ionic conductivity at RT (S cm ⁻¹)	T^+	Electrochemical window (V)
LiBAMB-PETMP/GBL (this work)	1.47×10^{-3}	0.89	5.2
PEGDM-LiMTFSI [52]	1.2×10^{-4}	0.86	5.5
PVDF-HFP/SG [51]	7.2×10^{-4}	0.89	4.1
PLDB@PVDF-HFP [54]	1.8×10^{-4}	0.89	5.0
PAE-LiPFS [19]	3.1×10^{-3}	0.98	4.7
CGPE [59]	2.29×10^{-4}	0.94	4.5
PVDF-HFP/LiPSIPA [50]	3.7×10^{-4}	0.82	5.8
PMMA-IL-TFSI/LiTFSI [21]	3.13×10^{-3}	0.96	5.18
PVDF-HFP/PTF-Li [25]	6.3×10^{-4}	0.86	4.3
LiPHFE [48]	4.1×10^{-4}	0.93	4.8
MTF-Li/PVDF-HFP [60]	6×10^{-4}	0.88	—

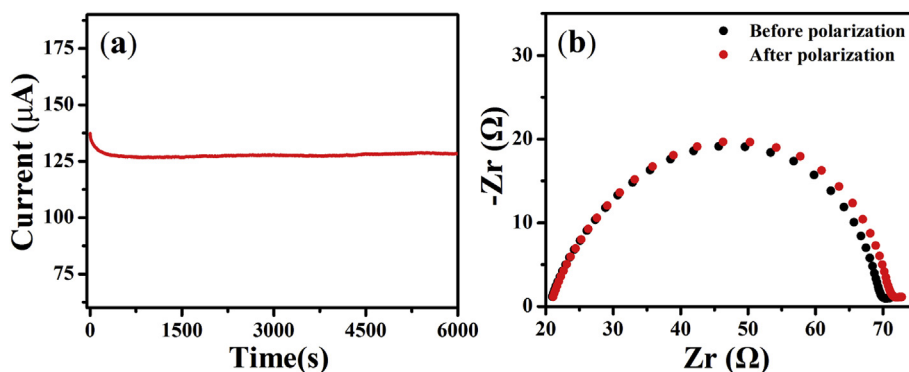


Fig. 5. (a) The time-dependent response of DC polarization for the Li|LiBAMB-PETMP/85wt. % GBL|Li symmetric cell. (b) Nyquist plots of the Li|LiBAMB-PETMP/85 wt. % GBL|Li symmetric cell before/after polarization at room temperature.

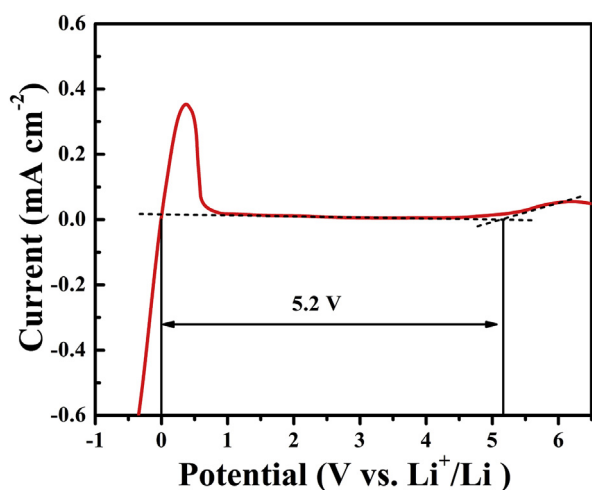


Fig. 6. Linear sweep voltammogram of the Li|LiBAMB-PETMP/85 wt. % GBL|SS cell from -1 V and 6.5 V (vs. Li^+/Li) at a scan rate of 1 mV s^{-1} .

3.4. Electrochemical stability

To evaluate the electrochemical stability of the LiBAMB-PETMP/GBL SIPE, a linear sweep voltammetry (LSV) study was carried out at room temperature between -1 V and 6.5 V (versus Li^+/Li) at a scan rate of 1 mV s^{-1} and the response is plotted as a current vs.

voltage curve, as depicted in Fig. 6. The scans show peaks between -0.5 and 0.6 V versus Li^+/Li that are associated with reversible lithium plating and stripping onto/from the working electrode [52]. There is no significant increase in the oxidation current up to 5.2 V, indicating that the LiBAMB-PETMP/GBL SIPE is electrochemically well suited for lithium ion battery applications.

3.5. Battery performance

To further demonstrate the promising prospects of the LiBAMB-PETMP/GBL SIPEs as polymer electrolytes in lithium ion batteries, LiBAMB-PETMP/85 wt. % GBL was used to assemble coin cells, with LiFePO_4 as the cathode and a lithium foil as the anode. The LiBAMB-PETMP/85 wt. % GBL film acts as both separator and electrolyte in the battery. The rate capability of the battery at various C-rates was examined at room temperature as shown in Fig. 7a. It is able to deliver of a specific capacity as high as 155 mA h g^{-1} at 0.2 C , which amounts to 91% of the theoretical capacity of LiFePO_4 . At higher current rates, it still exhibits high capacities of 146 mA h g^{-1} at 0.5 C , 136 mA h g^{-1} at 1 C and 123 mA h g^{-1} at 2 C . Even at a relatively high charging rate of 5 C , a stable capacity is obtained at 86 mA h g^{-1} . When the discharge/charge rate was switched from 5 C back to 0.2 C , the capacity recovered to 155 mA h g^{-1} , demonstrating highly reversible rate capability. To our knowledge, these values are remarkable higher than those of SIPE-based Li/LFP batteries disclosed to date. For example, a more recent work claimed that the discharge capacity of 150 mA h g^{-1} at 0.2 C is the highest of SIPE-

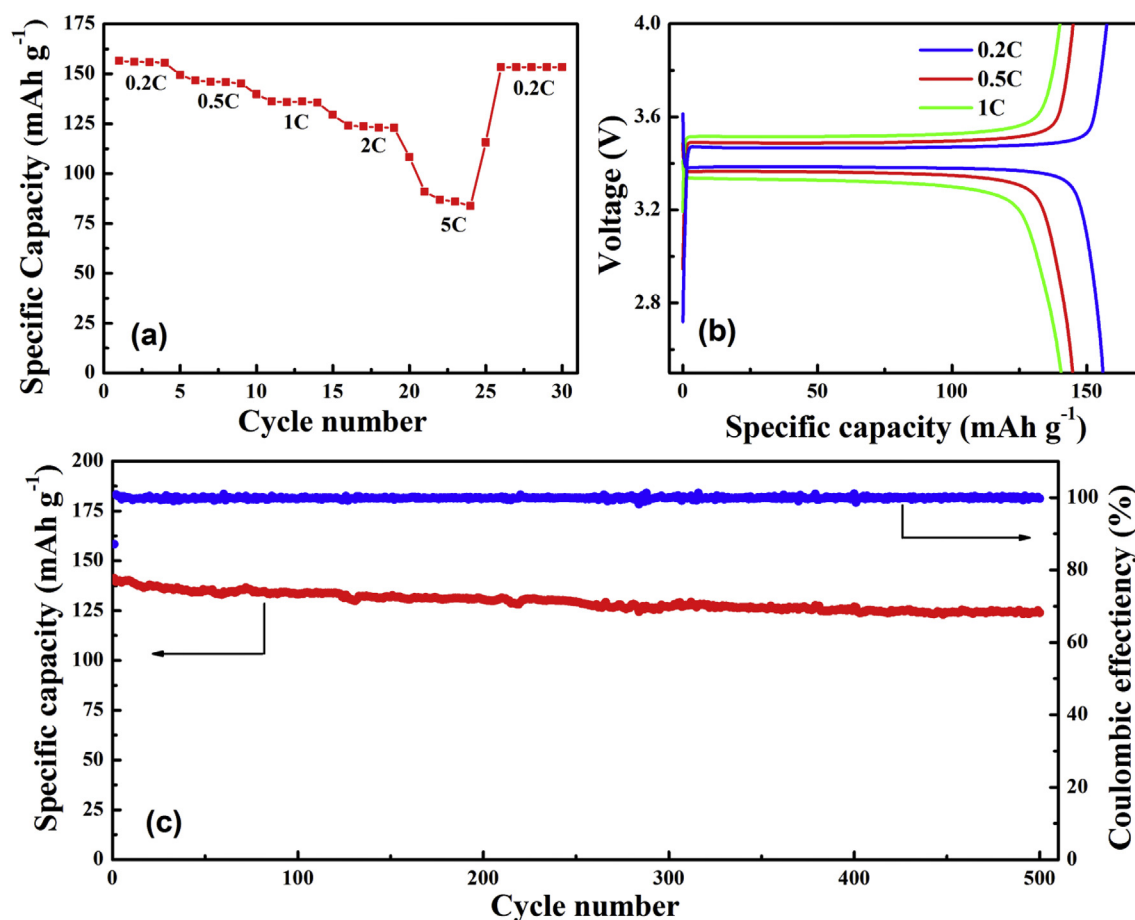


Fig. 7. Battery performance of Li|LiBAMB-PETMP/85wt. % GBL|LiFePO₄ cells at room temperature: (a) Rate capability; (b) Voltage plateaus at different C-rates; (c) Cycle performance at 1 C under room temperature.

based batteries reported to date [50]. The superior performance of the LiBAMB-PETMP/GBL LFP cell is mainly attributed to the high ionic conductivity and lithium ion transference number of SIPE and the low interfacial resistance between the membrane and the electrodes. Fig. 7b shows the discharge/charge curves of the LiBAMB-PETMP/GBL LFP cell at the rate of 0.2 C, 0.5 C and 1 C, all of which display typical discharge/charge plateaus of the Li/LFP cell at around 3.40 V.

Cycling performance of the battery is shown in Fig. 7c. Remarkably, the capacity remains at 124 mA h g⁻¹ at 1 C rate after 500 cycles. What's more, both capacity and the coulombic efficiency stay steady over all cycles, demonstrating its excellent long-term stability and reliability. It is an important step toward practical application of single-ion conducting polymer electrolyte in lithium ion batteries.

4. Conclusions

In summary, a network type sp³ boron-based single-ion conducting electrolyte can be readily synthesized by coupling lithium bis(allylmalonato)borate (LiBAMB) and pentaerythritol tetrakis(2-mercaptoacetate) via a one-step photoinitiated in situ thiol-ene click reaction in plasticizers. The ionic conductivities were found to vary with different kinds and contents of plasticizers. The SIPE containing 85 wt. % GBL as plasticizer (LiBAMB-PETMP/85 wt. % GBL) exhibits the highest ionic conductivity (1.47×10^{-3} S cm⁻¹ at 25 °C) and high lithium ion transference number of 0.89. This ionic conductivity is among the highest ionic conductivity of SIPEs reported to date. The significantly high value of lithium ion transference number and ionic conductivity is due to the restricted movement of anions that is part of crosslinking network, the weak coordination between lithium ions and sp³ boron anions as well as the improved lithium ions dissociation and mobility with the aid of GBL. LiBAMB-PETMP/GBL SIPEs shows electrochemical stability up to 5.2 V. More importantly, Li/LiFePO₄ cells with LiBAMB-PETMP/85 wt. % GBL as the electrolyte as well as the separator demonstrate highly reversible capacity and excellent rate capacity at room temperature, together with excellent long-term stability and reliability. The Li/LiFePO₄ cells maintain a capacity of 124 mA h g⁻¹ at 1 C rate even after 500 cycles without obvious decay. All these implies the promising prospects of the LiBAMB-PETMP/GBL SIPEs as polymer electrolytes in li-ion batteries, and it is an important step toward a practical development of SIPEs for high capacity, long-term stable and environmentally friendly lithium ion batteries.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jpowsour.2017.06.006>.

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